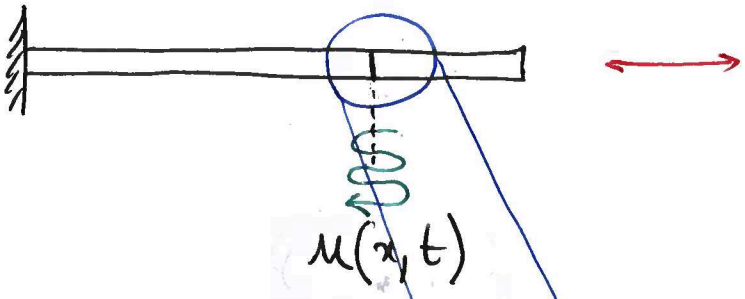
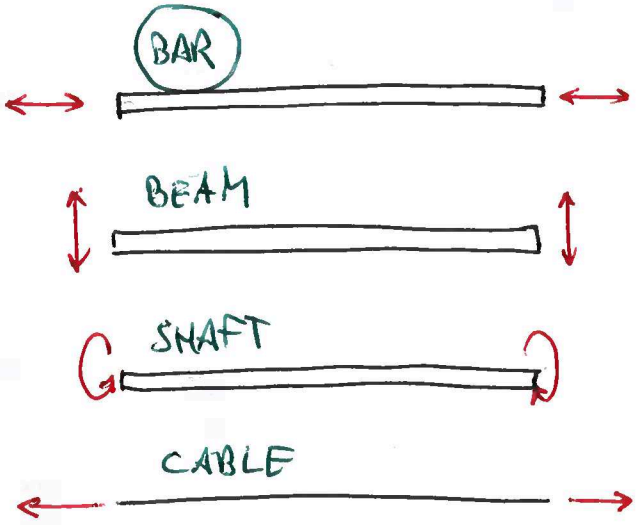
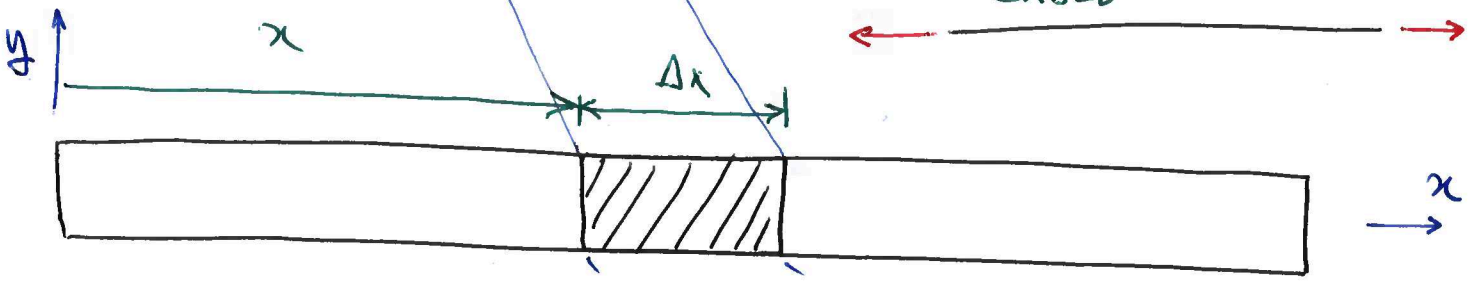
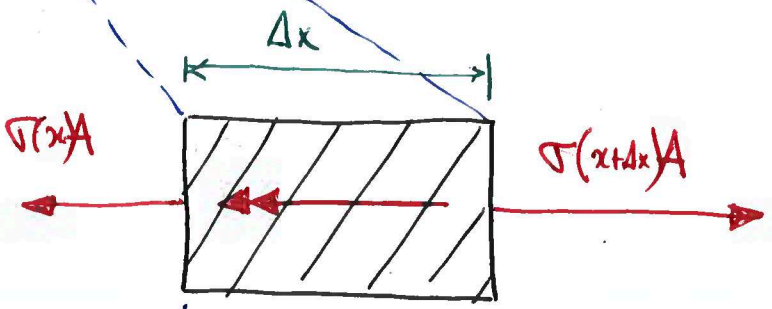
LM5

Lecture 14
5-Nov-2026





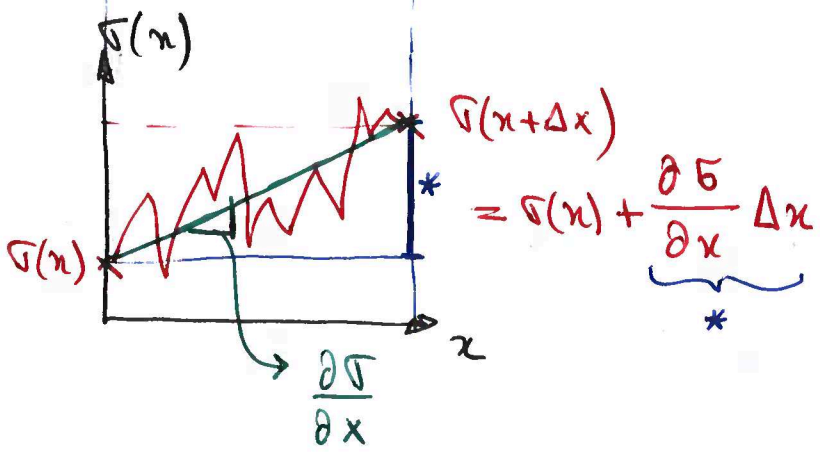




$$-\sigma(x)A + \sigma(x+\Delta x)A - \rho A \Delta x \ddot{u} = 0$$

$$\sigma(x+\Delta x) - \sigma(x) = \rho \Delta x \ddot{u}$$

?



$$\sigma(x) + \frac{\partial \sigma}{\partial x}\Delta x - \sigma(x) = \rho \Delta x \ddot{u}$$

$$\frac{\partial \sigma}{\partial x}\Delta x = \rho \Delta x \ddot{u}$$

$$\frac{\partial \sigma}{\partial x} = \rho \ddot{u}$$

$$\sigma = E \frac{\partial u}{\partial x}$$

$$\frac{\partial \sigma}{\partial x} = E \frac{\partial^2 u}{\partial x^2}$$

$$E \frac{\partial^2 u}{\partial x^2} = \rho \frac{\partial^2 u}{\partial t^2}$$

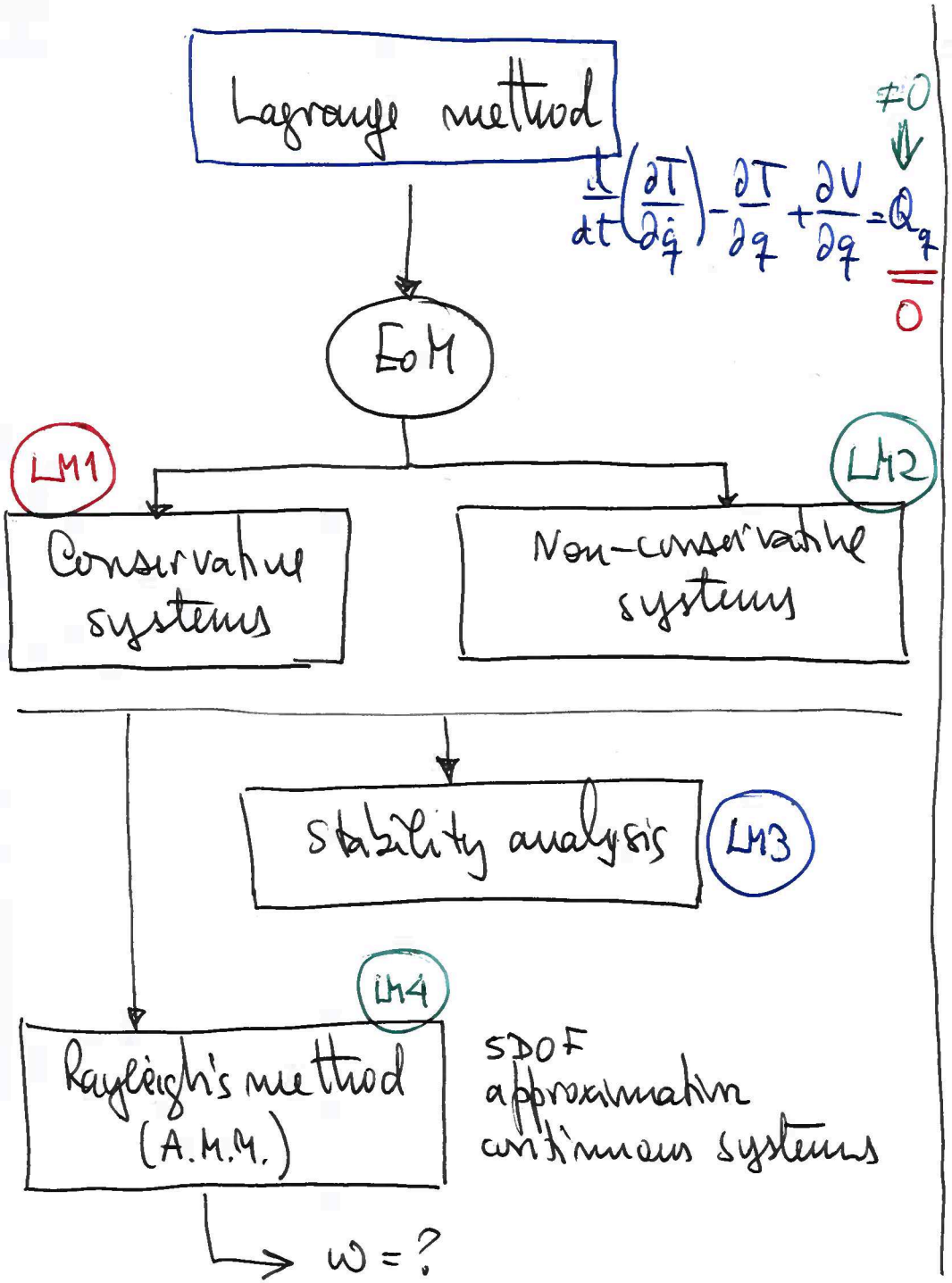
wave equation

$$F = k x$$

$$\sigma = E \varepsilon$$

Hooke's law
(Linear elastic)

$$\bar{\varepsilon} = \frac{\Delta x}{x} = \frac{\Delta l}{l} \qquad \varepsilon = \frac{\partial u}{\partial x}$$


Lagrange method
$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}}\right) - \frac{\partial T}{\partial q} + \frac{\partial V}{\partial q} = Q_q$
$\neq 0$
$=$
0
EoM
LM1
LM2
Conservative systems
Non-conservative systems
Stability analysis
LM3
LM4
Rayleigh's method (A.M.M.)
SDOF approximation continuous systems
$\omega = ?$